Chi-Square for Goodness of fit

In 2010 Census of the city, the weights of Individuals in a small city were found to be following

2010

< 50Kg	50-75	> 75
20%	30%	50%

Expected

In 2020 weight of n = 500 Individuals were Sampled. Below the results

< 50	50-75	> 75
140	160	200

using $\alpha = 0.05$, would you Conclude the population differences of weights has changed in the last 10 years.

2010 →

< 50Kg	50-75	> 75
20%	30%	50%

→ Expected

2020 →

< 50	50-75	> 75
140	160	200

Observed

n = 500

	< 50	50-75	> 75	
Expected	0.2×500 = 100	0.3×500 = 150	0.5×500 = 250	Exp

① Null hypothesis H_0 :- The data meets the Expectation

Alternate hypothesis H_1 :- The data does not meet the Expectation

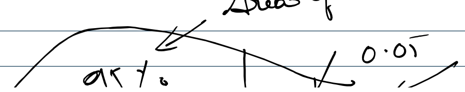
$\alpha = 0.05$ C I → 95%

③ Degree of freedom = K − 1 = 3 − 1 = 2

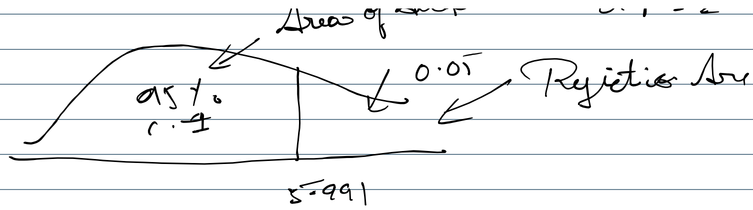
④ Decision Boundary

$\alpha = 0.05$
C.I = 95%
d.F = 2

table



Chi - Square table



If x^2 is greater than 5.991 we will reject null hypotheses else we fail to reject null hypothes

(5) Calculate chi Square test

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

2020 →

<50	50-75	>75	
140	160	200	Observ

n = 500

$$= \frac{(140-100)^2}{100} + \frac{(160-150)^2}{150} + \frac{(200-250)^2}{250}$$

	<50	50-75	>75	
Expected	0.2×500 = 100	0.3×500 = 150	0.5×500 = 250	Exp

$$= \frac{1600}{100} + \frac{100}{150} + \frac{2500}{250}$$

$= 16 + 0.66 + 10$

$= 26.66$

$x^2 = 26.666$

$x^2 = 26.66$ is greater than 5.991 so we will reject null hypotheses H_0

Conclusion / Inference is ⇒ The data does not meet the Expectation